Comparative Genomics in JBrowse 2 at the Rat Genome Database

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https://rgd.mcw.edu

Abstract

The Rat Genome Database (RGD; rgd.mcw.edu) is a multispecies knowledgebase that integrates genetic, genomic, and physiologic data. RGD has deployed JBrowse 2, a genome browsing tool that supports visualization and analysis for comparative genomics, including the ability to view multiple assemblies in a single session and dedicated views of structural variation and synteny. Here we describe an example use case illustrating how RGD's JBrowse 2 can be used for cross-species comparisons using Adcy3, a gene associated with obesity in rats, humans, and mice. The workflow demonstrates how to access JBrowse 2 from within the RGD site, open a region of interest, add variant and ortholog tracks, and utilize the linear synteny views. Starting from the RGD rat gene report page for Adcy3, a user can open a new JBrowse 2 window with the position of the Adcy3 gene automatically loaded. Adding ortholog tracks places the human and mouse gene models alongside the rat reference, with color indicating relative orientation. Variant tracks then layer strain-specific damaging variants and phenotypic allele variants associated with mutant rat strains documented in RGD. A parallel linear genome view can be added to show the human genes and transcripts track, along with other tracks such as the ClinVar pathogenic or likely pathogenic variants. The linear synteny view aligns the rat, human, and mouse genomes to show conservation of gene order and the surrounding context. Here, users can see the human block is reversed relative to rat and mouse, as evidenced by gene order and the crossed connections joining the rows. These views let users compare rat, human, and mouse data in a single session, visualizing variants and confirming how the surrounding regions align.

Access JBrowse 2 from RGD (rgd.mcw.edu)



There are multiple ways to access JBrowse from the RGD website:

1) Homepage → "Analysis and Visualization" → JBrowse button

2) Top menu → "Analysis and Visualization" tab → "JBrowse (Genome Browser)"

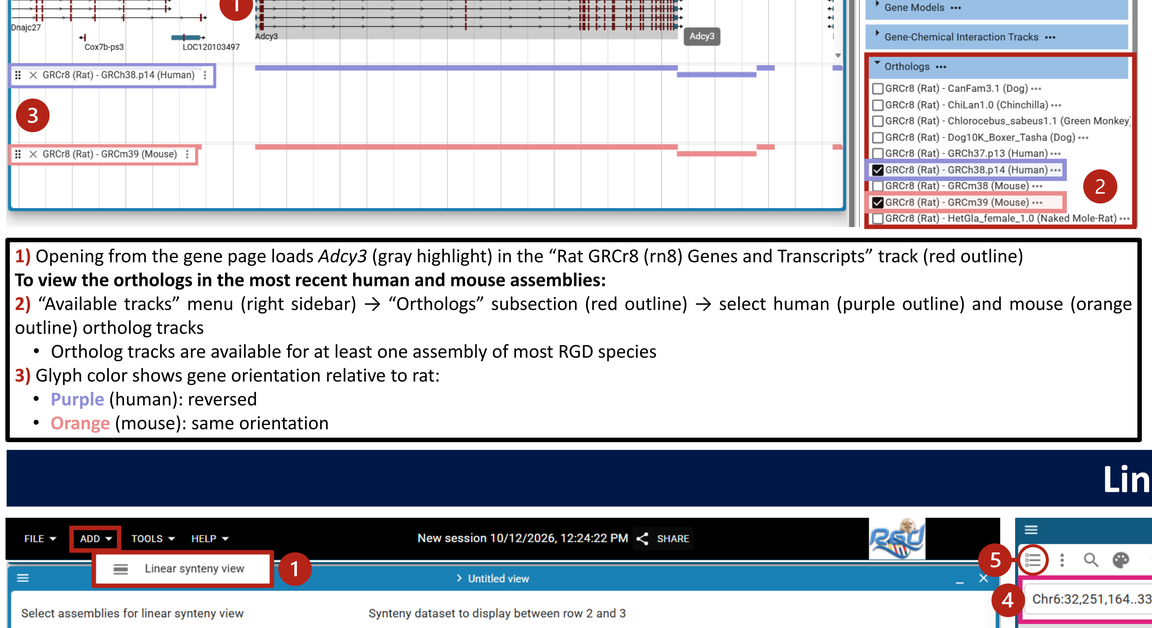
- Options 1 and 2 open a page where you can enter JBrowse 1 or JBrowse 2 by selecting the species assembly, which opens to a default view in JBrowse

3) From a gene page (example: *Adcy3*): search for a gene using the search bar

3a) Choose the number in the "Gene" row for the species of interest (shown here for rat)

3b) Select the gene from the search results table on the next page

3c) On the gene page, go to the "Position" section → "Genome Browsers" → "JBrowse" → linked assembly (GRCr8 for rat). This will automatically open the position of the gene in JBrowse 2



Ortholog Tracks

1) Opening from the gene page loads *Adcy3* (gray highlight) in the "Rat GRCr8 (rn8) Genes and Transcripts" track (red outline)

To view the orthologs in the most recent human and mouse assemblies:

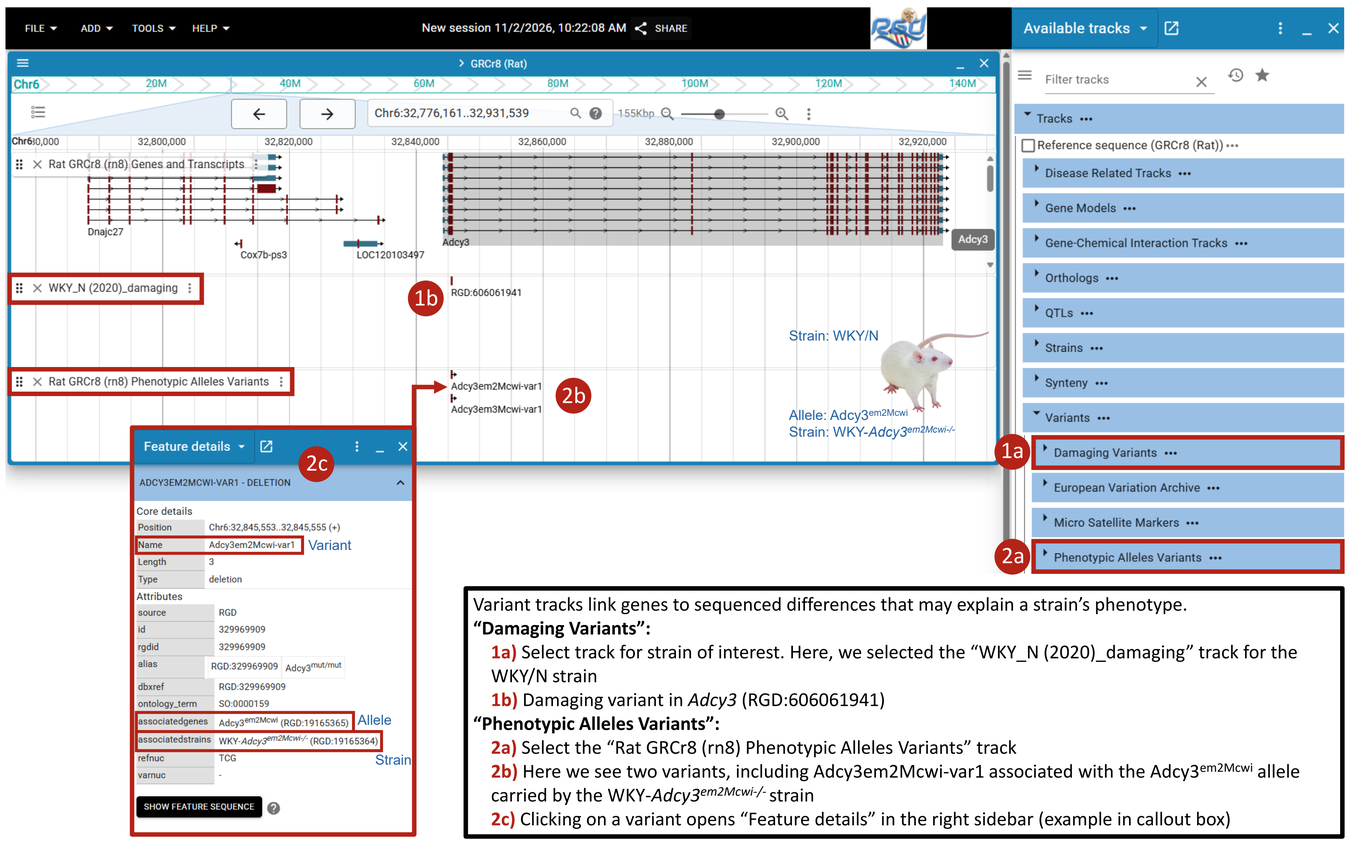
2) "Available tracks" menu (right sidebar) → "Orthologs" subsection (red outline) → select human (purple outline) and mouse (orange outline) ortholog tracks

- Ortholog tracks are available for at least one assembly of most RGD species

3) Glyph color shows gene orientation relative to rat:

- Purple (human): reversed
- Orange (mouse): same orientation

Rat Variant Tracks



Variant tracks link genes to sequenced differences that may explain a strain's phenotype.

"Damaging Variants":

1a) Select track for strain of interest. Here, we selected the "WKY_N (2020)_damaging" track for the WKY/N strain

1b) Damaging variant in *Adcy3* (RGD:606061941)

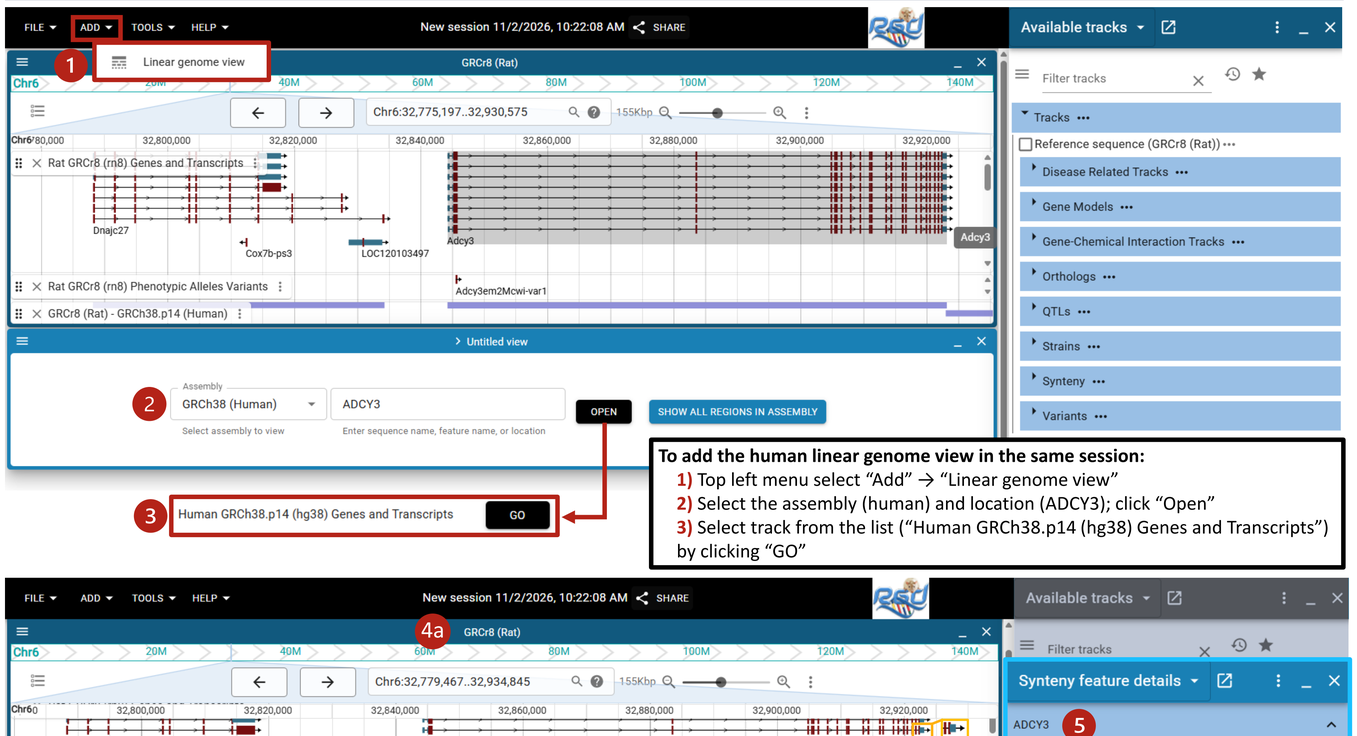
"Phenotypic Alleles Variants":

2a) Select the "Rat GRCr8 (rn8) Phenotypic Alleles Variants" track

2b) Here we see two variants, including Adcy3em2Mcwi-var1 associated with the Adcy3^{em2Mcwi} allele carried by the WKY-*Adcy3*$^{em2Mcwi-/-}$ strain

2c) Clicking on a variant opens "Feature details" in the right sidebar (example in callout box)

Human Linear Genome View and Variant Tracks

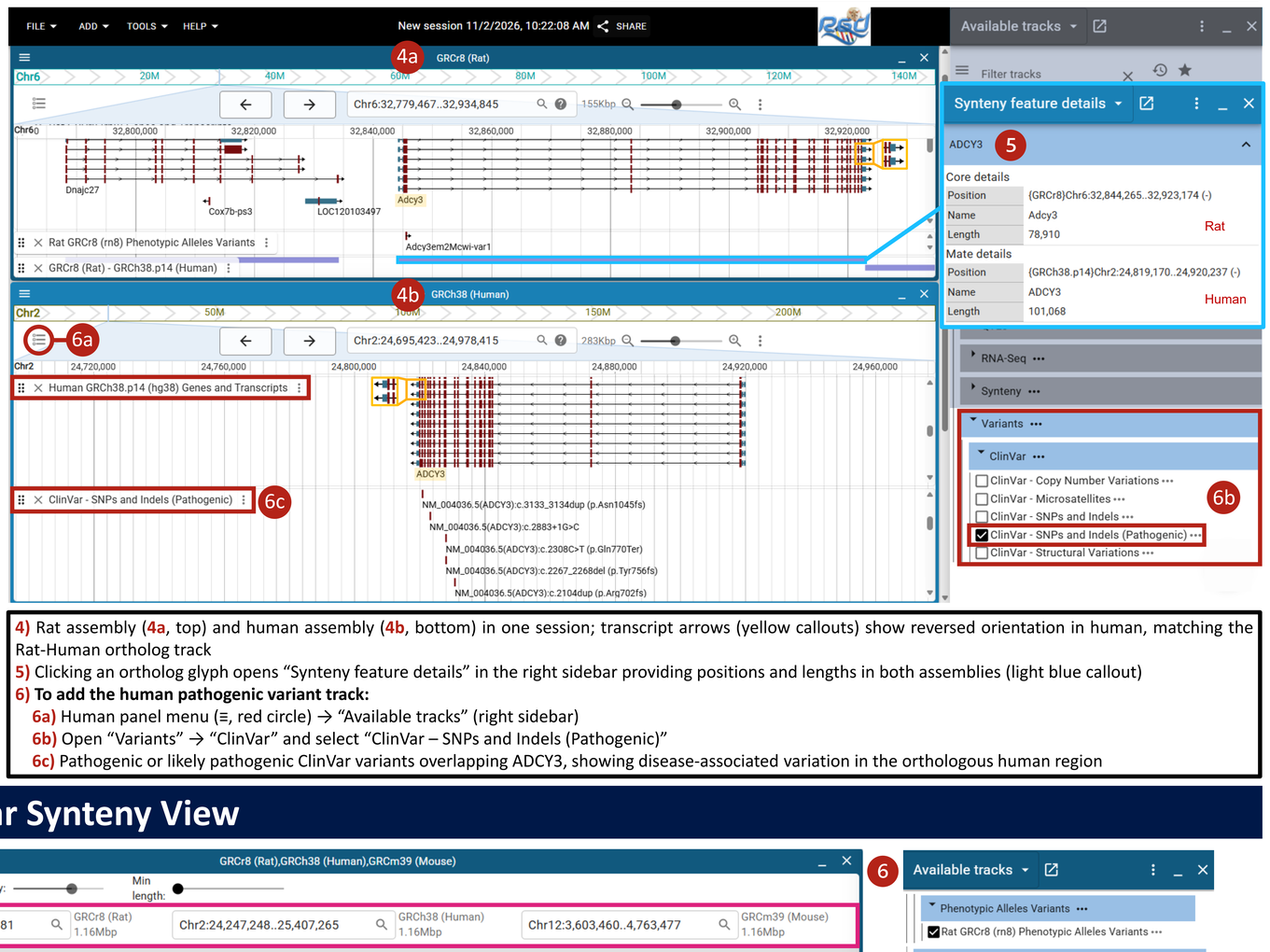


To add the human linear genome view in the same session:

1) Top left menu select "Add" → "Linear genome view"

2) Select the assembly (human) and location (ADCY3); click "Open"

3) Select track from the list ("Human GRCh38.p14 (hg38) Genes and Transcripts") by clicking "GO"



4) Rat assembly (**4a**, top) and human assembly (**4b**, bottom) in one session; transcript arrows (yellow callouts) show reversed orientation in human, matching the Rat-Human ortholog track

5) Clicking an ortholog glyph opens "Synteny feature details" in the right sidebar providing positions and lengths in both assemblies (light blue callout)

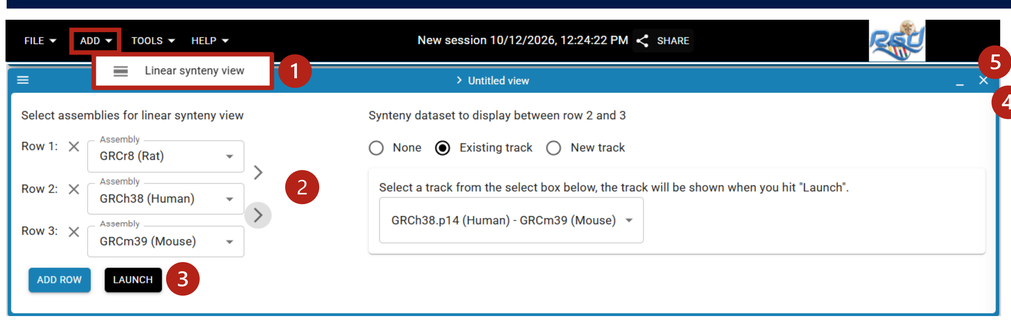
6) **To add the human pathogenic variant track:**

6a) Human panel menu (≡, red circle) → "Available tracks" (right sidebar)

6b) Open "Variants" → "ClinVar" and select "ClinVar – SNPs and Indels (Pathogenic)"

6c) Pathogenic or likely pathogenic ClinVar variants overlapping ADCY3, showing disease-associated variation in the orthologous human region

Linear Synteny View



To explore synteny across multiple species:

1) Top left menu select "Add" → "Linear synteny view"

2) Choose a species for each row (e.g., rat, human, mouse)

3) "Launch" displays genome-wide synteny blocks connected with red lines (initial display not shown)

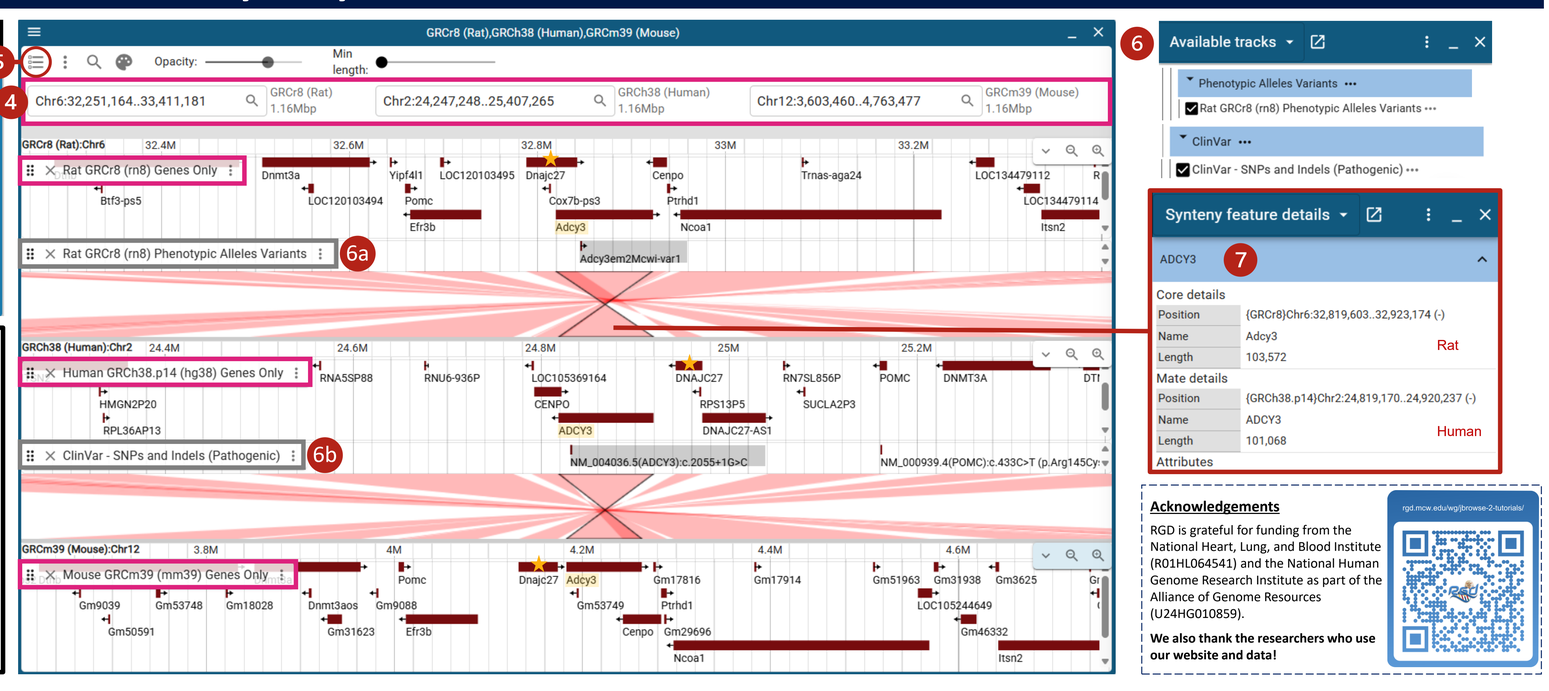
4) Search "Adcy3" in each location box (top pink outline); select the NCBI "genes only" track for each species (tracks outlined in pink; *Adcy3* highlighted in yellow). This view shows a single bar glyph for each gene, consolidating these tracks. The human block is reversed relative to rat and mouse, as shown by gene order (*Dnajc27* starred for reference) and by the distinctive (black) **X** pattern of the lines in the synteny tracks that join the orthologs between the genome assembly tracks

5) Additional tracks can be added to any or all of the syntenic regions shown. To bring up the "Available tracks" menu in the right sidebar (**6**), select the menu icon in the top left (≡, red circle) → "Row track selectors" → select row

6a) Select "Rat GRCr8 (rn8) Phenotypic Alleles Variants" and

6b) "ClinVar – SNPs and Indels (Pathogenic)" (grey highlights), the same tracks added in the previous figures

7) Clicking a synteny link opens "Synteny feature details" in the right sidebar (red callout)



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rgd.mcw.edu/wg/jbrowse-2-tutorials/